JOHNS HOPKINS
BLOOMBERG SCHOOL
*of* PUBLIC HEALTH

# Bias Control: Randomization and Masking

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# Bias

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# Random and Systematic Error

Measured value = True value + Error

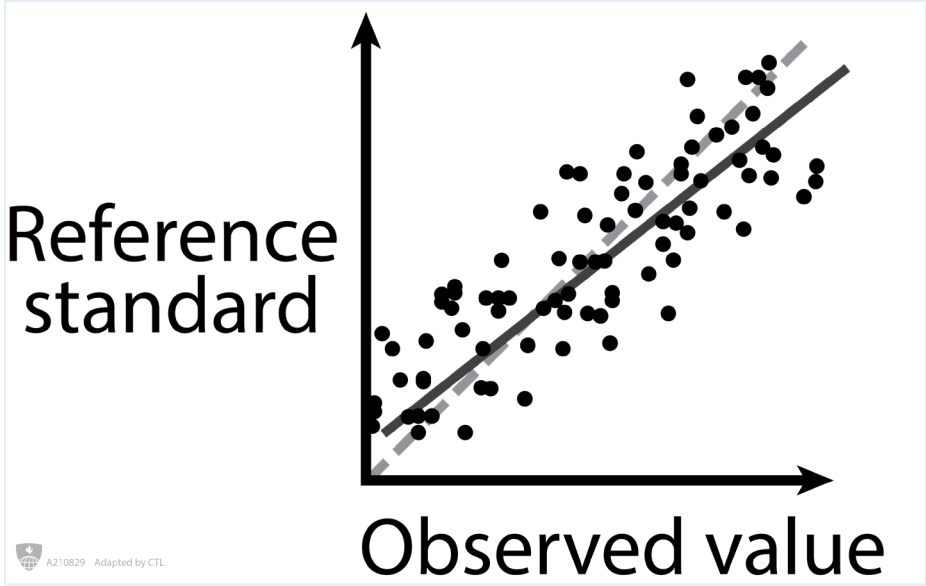
Error = Bias + Random

Systematic component of the error

Unpredictable component of the error

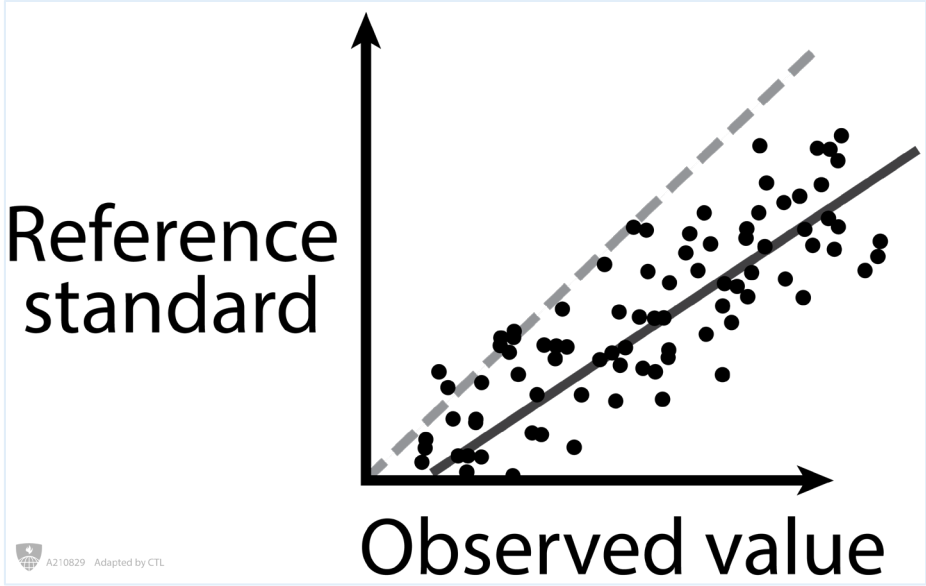
# Random Error

- Normally distributed around mean
- Mean of errors is zero
- May be reduced by study design



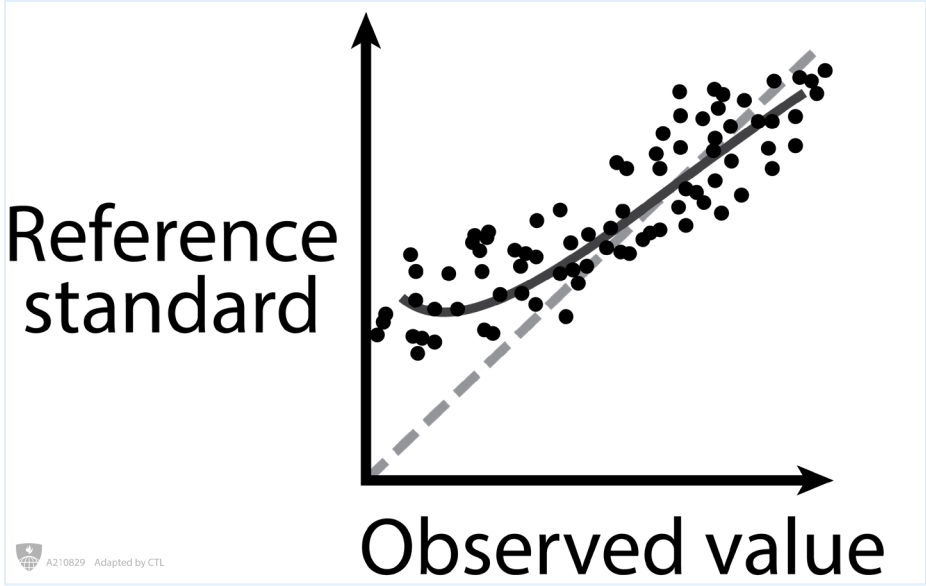
# Systematic Error (Bias)—1

- Measurements systematically distorted away from the true value
- Example: overestimation of physical activity when using certain questionnaires
- May be corrected in the analysis

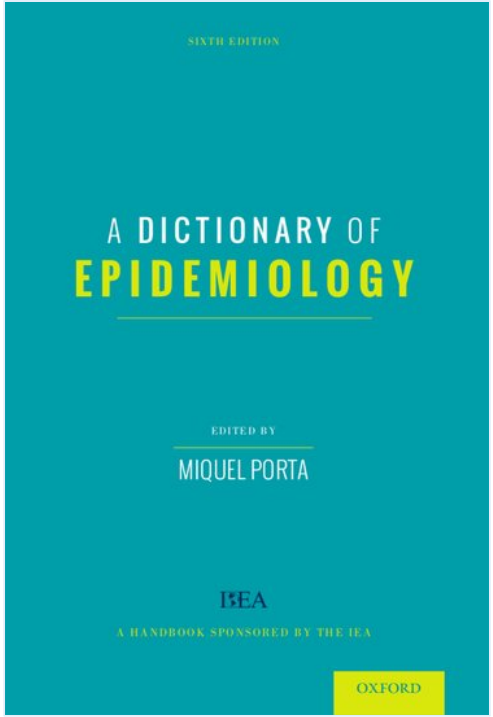


# Systematic Error (Bias)—2

- Here: error is very hard to predict
- Example: underestimation of physical activity at low levels when using pedometers



# Bias—One Definition




[Modified]: "Systematic (as opposed to random) deviation of results or inferences from the truth."

"The term *bias* does not necessarily carry an imputation of prejudice such as the experimenter's desire for a particular outcome. This differs from conventional usage, in which bias refers to a partisan point of view—to prejudice or unfairness."

Source: Porta, Miquel. (2014). *A dictionary of epidemiology.* (6th ed.) New York: Oxford University Press.

# Randomization

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# What Is *Randomization?*

- An explicit method that uses the play of chance to assign participants to comparison groups in a trial
- A process where treatment assignment follows no describable deterministic pattern but follows a probabilistic distribution

# Clinical Equipoise

*Prerequisite for Randomization*

"The randomized trial arises from a climate of clinical equipoise. Its ethical base rests on the presence of a *collective state of doubt* regarding the course of treatment, sufficiently balanced to justify randomization."

Source: Meinert, CL. (2012). *Clinical Trials: Design, Conduct and Analysis.* Volume 39 of Monographs in Epidemiology and Biostatistics, ISSN 0740-0845. Oxford University Press, USA.

# Why Do We Randomize?

- To minimize confounding for ***known and unknown*** factors that can cause the outcome

- To minimize selection bias:
  - By preferentially assigning participants to one group or another in a way that affects the outcome
  - Example: giving healthier (or sicker) patients the “better” treatment

# Effect of Randomization

- Produce comparable treatment groups
- Define entry into the trial
- Meet assumptions of statistical tests

# Does Randomization Eliminate all Confounding and Bias?

- Not all bias is minimized by randomization
  - Baseline differences?
  - Differences in drop-out proportions between groups?
  - Outcome measurement differences?

- Baseline differences—a “failure of randomization”?
  - Tables of “baseline characteristics” (“table 1”)
  - Show p-value?

# Real World Example: *Is this Randomization?*

Study evaluating a new procedure versus the old procedure in an intensive care unit (ICU) setting

"[…] During the day, the new procedure was usually used. At night, the physician was often without support staff and did not use the new procedure very often. The result was a ~ 1 : 1 randomization (new procedure : old procedure) […]"

Source: Email from a good friend of mine.

# Types of Randomization

- Simple
- Restricted
- Adaptive

# Simple Randomization—1

- Each assignment has the same probability of selection
- Like flipping a coin

**Advantages**

- Simple
- Easy to deploy
- Unpredictable

# Simple Randomization—2

- Each assignment has the same probability of selection
- Like flipping a coin

| Advantages | Disadvantages |
| --- | --- |
| ▶ Simple<br>▶ Easy to deploy<br>▶ Unpredictable | ▶ Not reproducible<br>▶ Risk of imbalance (allocation ratio and baseline characteristics) |

# Simple Randomization—3

- Example: two groups, 1:1
- 100 assignments
- Use Excel random numbers
- Out of 50 tests
  - Only two ended at 50/50
  - Four ended with 60+ in one group
  - Max: 68/32

# Restricted Randomization: *Blocking—1*

- At least some assignments are determined from prior assignments
- Example: block of four—AABB, ABAB, ABBA, BBAA, BABA, BAAB

| Advantages | Disadvantages |
| --- | --- |
| ▶ Better balance (allocation ratio) | ▶ More predictable |

# Restricted Randomization: *Blocking*—2

- In practice, tables are often composed of a mix of block sizes
  - I.e., mix of blocks of size two and four

- Example: AABB BABA AB BAAB BA AB
  - Three blocks of four
  - Three blocks of two

- More difficult to predict

- Blocking scheme (size and number of blocks) should be kept confidential or changed randomly

# Restricted Randomization: *Blocking*—3

| Advantages of blocking | Disadvantages of blocking |
| --- | --- |
| ▶ Imbalance minimized<br>▶ Complete balance achieved periodically<br>▶ Guards against temporal changes (i.e., seasonality) | ▶ More complex to construct<br>▶ Some assignments predictable |

# Restricted Randomization: *Blocking*—4

- Smallest possible block size is the sum of the factors in the allocation ratio
  - 1:1
    - Smallest block is two
  - 1:2:1
    - Smallest block is four
  - 1:2:3
    - Smallest block is six

- Larger blocks should be multiples of the smallest possible block size
  - 1:2:3 could have blocks of size six and twelve

# Restricted Randomization: *Stratification*—1

- ***Stratified assignment:*** subgrouping assignments based on characteristics measured *before* assignment
- Usually, characteristics related to the outcome of interest
- Common: gender, age group, study site (in a multicenter trial)
- Create a separate assignment schedule for each value of each stratification variable

# Restricted Randomization: *Stratification*—2

- Example: a multicenter study with 10 sites that stratifies by site will have 10 separate assignment schedules
  - One for each site
- If that same study also wants to stratify by age group (adult versus pediatric), it will need 20 separate assignment schedules
- Each assignment schedule is created independently

# Restricted Randomization: *Stratification*—3

- Attempt to achieve comparability between groups regarding important prognostic factors (stratification variables)
- Typically used together with blocking
- Minimize the number of strata (they might get small; imbalances might result)

# Adaptive Randomization: Two Examples

| 1. Response-adaptive randomization | 2. Covariate-adaptive randomization |
| --- | --- |
| ► Uses interim data to unbalance the allocation probabilities in favor of the "better" treatment(s)<br>► "Playing the winner" | ► Can balance the group composition with respect to the most important covariates<br>► E.g., age, biomarkers |

# Practical Tips for Randomization

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# Good Randomization Practices

- Document the assignment schedule
- Collect all baseline data ***before*** assignment
- Participant should be eligible and ready to begin
- Assignments are irrevocable
  - No exceptions to order of assignments
- Assignments are concealed until used (“concealment of random allocation”)
- Usually combined with treatment masking

# Avoid Quasi-Randomization

- Day of the week
- Birth date
- Alphabet

- Not random
- Predictable assignment

# Practical Aspects of Issuing Treatment Assignment

| Envelopes | Telephone | Online |
| --- | --- | --- |
| ▶ Simple<br>▶ Subject to misconduct | ▶ Cumbersome | ▶ Expertise required<br>▶ Feasible and accepted |

# Primer: Intention-to-Treat Analysis Versus Per-Protocol Analysis

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# Intention-to-Treat (ITT) Analysis

- Everyone in a trial is analyzed in their assigned groups, irrespective of what happens after randomization
- "Analyze as randomized"
- Even if participants:
  - Have been falsely included (did not meet inclusion criteria)
  - Did not receive any treatment
  - Did not receive treatment to which they have been randomized

# Per-Protocol (PP) Analysis

- Participants are analyzed “as treated”
- Sometimes called “treatment received”
- Problems with this approach:
  - Violates randomization
  - Crossovers are often not random

# ITT Versus PP (in Superiority Trials)

- While ITT is very conservative, it is the standard primary analysis in randomized controlled trials (RCTs)

- PP is typically also performed, but as a:
  - Secondary analysis
  - Safety analysis
  - To complement the ITT analysis

# Other Ways to Address Confounding in Clinical Trials

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# Restriction to Control for Confounding

- Only include people with one level of the confounder into your trial
- Example: if age is an important confounder, you may include only people of a certain age range
- Disadvantage: comes at the price of reduced generalizability

# Adjusting for *Baseline Variables* to Control for Confounding

- Not often done in trials but is a powerful means to reduce confounding
- May increase the statistical precision of the trial
- Only use variables that are present and measured ***before randomization***
  - Such as age, some lab value, comorbidities
- Important to not “disturb” the randomness after randomization

# Masking

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# About Masking

| What is *masking?* | What do we typically mask? |
| --- | --- |
| ▶ Concealing something from:<br>▶ Study team<br>▶ Participant<br>▶ Outcome assessors<br>▶ Others<br><br>▶ Sometimes called *blinding*<br>▶ Synonymous | ▶ Treatment<br><br>▶ Outcome |

# Why Mask?

Reduces bias (information bias)

- Participants
  - Self-reported data
  - Placebo effect
- Investigators
  - Data collection
  - Outcome assessment
- Adverse event assessment

# Who Can Be Masked?

| One group (usually participant) | Both participant and investigator | Includes an independent outcome assessor | Includes statistician |
| --- | --- | --- | --- |
| ▶ Single masked | ▶ Double masked | ▶ Triple masked | ▶ Quadruple masked |

# Masking: Terminology

- The terminology is inconsistent!
- When reading reports check who ***exactly*** has been masked

# Masking Can Be Difficult

| Expensive to mask drug | Often poorly done | Often easy to unmask |
| --- | --- | --- |
| ▶ Placebo often more expensive than active drug | ▶ Weight, color, appearance, texture, taste, etc. | ▶ Side effects<br>▶ Open capsules<br>▶ Weigh packages |

# Masking Can Be Impossible

- Some surgical trials
- Behavioral interventions

- Rule of thumb: even if you can’t mask everyone, you should still mask as many people as you can (reduce information bias)

# Summary

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# Randomization and Masking

| Randomization | Masking |
| --- | --- |
| ▶ Does not guarantee comparable groups<br>▶ But it does help<br>▶ Differences at baseline do not mean randomization failed<br><br>▶ Helps address confounding and selection bias<br><br>▶ Helps comparability for measured and unmeasured characteristics | ▶ Another bias control element (information bias) |